Multilevel Image Thresholding for Brain Tumor Segmentation using Quantum-Inspired Grasshopper Optimization Algorithm

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Abstract—Accurate segmentation of brain tumors from Magnetic Resonance Imaging (MRI) is a critical task in medical diagnosis. Multilevel thresholding is an effective technique for image segmentation, but its computational complexity increases exponentially with the number of thresholds. This paper proposes a Quantum-Inspired Grasshopper Optimization Algorithm (QIGOA) for optimal multilevel thresholding. By integrating quantum computing principles, such as Q-bits and quantum rotation gates, into the standard GOA, we enhance the global search capability and convergence speed. The proposed method utilizes Kapur's entropy as the objective function to determine the optimal thresholds. Experimental results on the BraTS dataset demonstrate that QIGOA outperforms state-of-the-art heuristic algorithms in terms of Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and CPU time.

Index Terms—Brain Tumor, MRI, Multilevel Thresholding, Grasshopper Optimization Algorithm, Quantum-Inspired Computing, Kapur's Entropy.

I. Introduction

Brain tumor identification requires high precision as any error can lead to fatal consequences. MRI is the most widely used imaging modality for this purpose. However, segmenting tumors is challenging due to complex shapes and blurred boundaries. Meta-heuristic algorithms have shown great potential in finding optimal thresholds for segmentation. Among them, the Grasshopper Optimization Algorithm (GOA) is inspired by the social interaction of grasshopper swarms. To further improve its robustness, we introduce a quantum-inspired version.

II. Problem Formulation

A. Multilevel Thresholding

Let I be a grayscale image with L intensity levels $\{0, 1, \ldots, L-1\}$. The goal is to find k thresholds $\{t_1, t_2, \ldots, t_k\}$ that maximize the Kapur's Entropy function:

$$f(t_1, t_2, \ldots, t_k) = H_0 + H_1 + \cdots + H_k \tag{1}$$

where H_j is the entropy of the j-th class.

B. Quantum Representation

In QIGOA, a quantum bit (Q-bit) is defined as $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. A grasshopper's position is represented by a Q-bit individual:

$$q_i = \begin{bmatrix} \alpha_{i1} & \alpha_{i2} & \ldots & \alpha_{in} \\ \beta_{i1} & \beta_{i2} & \ldots & \beta_{in} \end{bmatrix} \tag{2}$$

III. The Proposed QIGOA Algorithm

A. Standard Grasshopper Optimization Algorithm (GOA)

The position of the i-th grasshopper X_i is updated as:

$$X_i = c\left(\sum_{j=1, j\neq i}^{N} c\frac{ub-lb}{2} s(|x_j - x_i|)\frac{x_j - x_i}{d_{ij}}\right) + \hat{T} \tag{3}$$

where $\hat{T}$ is the best solution found so far (the target).

B. Quantum Rotation Gate Update

In our QIGOA, the update process is performed by a quantum rotation gate $U(\Delta\theta_i)$:

$$\begin{bmatrix} \alpha_i^{new} \\ \beta_i^{new} \end{bmatrix} = \begin{bmatrix} \cos(\Delta\theta_i) & -\sin(\Delta\theta_i) \\ \sin(\Delta\theta_i) & \cos(\Delta\theta_i) \end{bmatrix} \begin{bmatrix} \alpha_i^{old} \\ \beta_i^{old} \end{bmatrix} \tag{4}$$

The rotation angle $\Delta\theta_i$ is a function of the current fitness and the target fitness, ensuring the swarm moves towards the global optimum.

Algorithm 1 QIGOA for Multilevel Thresholding

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Load MRI image and compute histogram.
Initialize Q-bit population Q = {q_1, q_2, ..., q_N}.
Generate initial positions by observing Q-states.
Evaluate fitness using Kapur's Entropy.
Set T̂ = best grasshopper (optimal thresholds).
while l < Max_Iterations do
    Update c parameter (decreases linearly).
    for each grasshopper do
        Apply Quantum Rotation Gate based on T̂.
        Perform measurement to obtain discrete thresholds.
        Check boundary conditions.
    end for
    Update T̂ if a better fitness is found.
    l = l + 1
end while
return T̂
```

IV. Experimental Results

We utilized the BraTS 2023 dataset for validation. The performance is compared against standard GOA, Particle Swarm Optimization (PSO), and Genetic Algorithm (GA).

A. Quantitative Analysis

The metrics used for evaluation include:

- PSNR: Measures the quality of the segmented image.
- SSIM: Measures the structural similarity between the original and segmented images.
- Fitness Value: The maximum Kapur's entropy achieved.

TABLE I
Comparison of PSNR and CPU Time

Algorithm	Thresholds (k)	PSNR (dB)	CPU Time (s)
GA	5	18.45	2.14
PSO	5	19.12	1.85
GOA	5	20.34	1.56
QIGOA	5	22.87	0.92

V. Conclusion

This paper introduced a Quantum-Inspired Grasshopper Optimization Algorithm for brain tumor segmentation. By combining quantum computing dynamics with the swarm intelligence of grasshoppers, the proposed QIGOA effectively finds optimal thresholds in a high-dimensional search space. Future work will involve 3D MRI segmentation and deep learning integration.

References

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